

Option B – Medical Physics

7. (a) An X-ray machine is designed to fire enormous numbers of electrons per second at very high speeds into a metal target. To do this it typically operates with a current of 100 mA. The electrons are accelerated from rest through a distance of about 3 cm hitting the target at half the speed of light.

(i) State how fast moving electrons can produce X-rays. [1]

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(ii) Determine the number of electrons arriving at the target every second. [2]

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(iii) Calculate the acceleration of an electron in the X-ray tube as it travels towards the target. [3]

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(iv) Explain how the photon energy and the intensity of the X-ray output can be increased. [2]

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- (b) Give **one** advantage and **one** disadvantage of using CT scans compared with conventional X-ray imaging. [2]

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- (c) The following table provides information relating to ultrasound in fat and muscle.

Tissue	Density / kg m ⁻³	Speed / ms ⁻¹
Muscle	1.1×10^3	1.6×10^3
Fat	0.8×10^3	1.5×10^3

By considering acoustic impedance, explain whether or not you would expect to receive an ultrasound echo from a muscle-fat boundary. [4]

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- (d) (i) Explain briefly the role played by the absorbed radio waves in the resonance process of MRI scans. [2]

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- (ii) In an MRI scanner radio waves of frequency 64 MHz are absorbed and cause hydrogen atoms to resonate in a magnetic field of 1.5 T. Determine the frequency of absorbed radio waves that would be needed if the field was reduced to 1.2 T. [2]

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- (e) (i) Alpha particles have a radiation weighting factor of 20. If a patient receives an absorbed dose of 55 mGy determine the dose equivalent received. [1]

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- (ii) State whether or not beta particles would have a weighting factor of 20, greater than 20 or less than 20. Justify your answer. [1]

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